Problem statement : As a data scientist, you must develop a model which will predict the insurance charges

1. From given data insurance charges prediction is the clear ask from client. Hence Output is "charges"
2. Rows - 1338 and columns – 6 from the source file. 5 columns contribute to input and 1 column contribute to output.
3. Columns Sex & smoker are given with data in nominal – can be converted using **one-hot coding**

Since the Problem statement talks about numbers	Stage 1 . Machine learning domain
Since input and output are clearly mentioned	Stage 2 : Supervised Learning
Since input and output given are in numbers except the columns of Sex i.e male/female and Smoker - yes/no that can be converted into nominal data using **one-hot coding**	Stage 3 : Regression

Simple Linear Regression: Not a fit as inputs has multiple attributes

Multi Linear Regression: r_score : 0.795

SVM	**Before data standardization**	**RBF (R value)**	**Poly (R value)**	**Linear (R value)**	**Sigmoid(R value)**
	C=10	-0.095	-0.121	-0.041	-0.099
	C=100	-0.155	-0.131	0.521	-0.124
	C=500	-0.156	-0.116	0.613	-0.419
	C=1000	-0.149	-0.092	0.618	-1.521
	C=2000	-0.142	-0.042	0.625	Decreasing further
	C=3000	-0.139	0.006	0.666	Decreasing further

SVM	**After data standardization**	**RBF (R value)**	**Poly**	**Linear**	**Sigmoid**
	C=10	-0.048	0.027	0.432	0.019
	C=100	0.291	0.604	0.616	0.505
	C=500	0.639	0.815	0.681	0.463
	C=1000	0.791	0.851	0.759	0.184
	C=2000	0.846	0.857	0.761	Decreasing further
	C=3000	0.861	0.858	0.761	Decreasing further

Decision Tree	With default params	r_score =0.712		
S.NO	**Criterion**	**Max Features**	**Splitter**	**R- Value**
1	Squared error	None	best	0.693
2	Squared error	None	random	0.711
3	friedman_mse	None	best	0.743
4	friedman_mse	None	random	0.743
5	absolute_error	None	best	0.705
6	absolute_error	None	random	0.774

Random Forest	**Params**	**R- value**
1	n_estimators=50,random_state=0,criterion=squared_error	0.855
2	n_estimators=100,random_state=0,criterion=squared_error	0.857
3	n_estimators=100,random_state=0, criterion=absolute_error	0.856
4	n_estimators=50,random_state=0, criterion=absolute_error	0.853

Conclusion: SVMR after data standardization we can get the max r_score=0.861 and hence considering the same with kernel='rbf' & C=3000 as final model .